Miguel Shinyenyi

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PROFESSIONAL SUMMARY

Backend Software Engineer with 4+ years of experience building scalable, data-intensive systems in Java and Spring Boot. Proven track record delivering multi-tenant ERP platforms, RESTful microservices, and automated data pipelines processing hundreds of daily transactions across distributed systems. Strong foundation in event-driven architecture, PostgreSQL optimisation, TDD, and secure access control. Writes technical deep-dives on the systems he builds and is developing umwayi-os, an open-source rules and oversight system for AI helpers, as a working example of the same practices.

WORK EXPERIENCE

Software Engineer, Global Link+

Sep 2023 – Present
Utah, USA (Remote)

Multi-tenant ERP logistics platform serving 10 B2B clients, with integrations across Shopify, Amazon, and Walmart.

- Designed and maintained RESTful backend services in Java Spring Boot powering a multi-tenant ERP platform, enforcing strict data isolation across 10 B2B clients and their customer bases, processing hundreds of orders daily.
- Built and optimised backend APIs for a real-time admin dashboard tracking client shipment volumes, order activity, and logistics KPIs, combining live PostgreSQL queries with pre-aggregated data for historical reporting.
- Integrated Stripe for secure payment processing, improving transaction reliability and the client-facing payment experience.
- Implemented scheduled background jobs to automate client report generation, aggregating daily order and shipment data across 10 clients into structured summaries and eliminating manual reporting overhead.
- Introduced Redis caching for frequently-accessed, low-volatility dashboard data and optimised PostgreSQL queries through indexing and query planning, reducing API latency by 40% and cutting redundant database load under peak concurrent usage.
- Architected and delivered a secure file management system storing tens of thousands of documents (contracts, product images, shipping records) with role-based access control enforced at the API layer.
- Designed a multi-tenant PostgreSQL data model with tenant-scoped schemas and query filtering, ensuring data integrity and isolation across all client tiers without relying on application-layer safeguards alone.
- Contributed to CI/CD pipeline setup by writing Maven build scripts to automate compilation, testing, and packaging for consistent, repeatable deployments.
- Improved system observability by integrating JVM metrics and backend health checks into monitoring dashboards, surfacing performance bottlenecks before they reached clients.
- Participated in code reviews and pair programming, upholding clean-code principles and consistent engineering standards across the Java codebase.
- Collaborated with product and operations teams to translate logistics workflows into backend features, bridging business requirements and technical implementation in a cross-functional startup environment.

Software Engineer, IBM Research

Jun 2023 – Aug 2023
Nairobi, Kenya

Backend services supporting real-time molecular visualisation and AI-driven computation for scientific research workflows.

- Developed Flask-based APIs for real-time molecular visualisation and AI-driven computation, working directly with researchers to diagnose integration issues and improve data accuracy and latency.
- Built and tuned PostgreSQL queries for analytical workloads, applying indexing and query planning to support high-read scientific data pipelines.
- Containerised backend services using Docker, ensuring consistent deployment and environment reproducibility across research and production systems.

Software Engineer, Quikk API — Sep 2022 – May 2023

Nairobi, Kenya

Spring Boot microservices powering an analytics-driven API platform.

- Built and optimised Spring Boot microservices with JWT authentication, keyset pagination, and database indexing for secure, performant API access.
- Integrated PostHog for real-time user telemetry and performance metrics, supporting proactive troubleshooting and data-driven product decisions.
- Conducted unit and integration testing following TDD principles, maintaining high test coverage and contributing to service reliability.

PROJECTS & WRITING

umwayi-os, open-source rules system for AI helpers (in development) — github.com/Miguel-Shinyenyi/umwayi-os

- Building a command-line system in Python, with no third-party packages in the core, that checks every action an AI helper wants to take against a rules file and answers Yes, No or Ask me first. Unknown actions default to Ask me first, and any internal error blocks the action.
- Implemented eight layers: memory, rules check, approval queue, archive, tamper-evident logbook, scheduled jobs with an allow list, read-only helpers that cannot approve their own work, and workflows with a kill switch.
- Directed an AI coding agent to build each layer from a written specification, then reviewed, tested and merged it by pull request. More than 500 automated tests, continuous integration on Python 3.10 and 3.12, and a security scan before each merge.

umwayi-rm, savings group assistant (Meta hackathon) — github.com/Miguel-Shinyenyi/umwayi-rm

- An AI helper proposes ledger entries for a savings group and a human treasurer approves them. No money moves. Wrote the test plan before running experiments; both main hypotheses were not supported, and I reported that.

Invoice-Financing, open-source settlement & reconciliation engine — github.com/Miguel-Shinyenyi

- Personal fintech backend project: an idempotent settlement engine with a transactional outbox, double-entry ledger consistency checks, and concurrency-safe idempotency-key handling under load.

Technical writing, personal site, dev.to, Substack — miguel-shinyenyi.github.io/miguel-site

- Writes and publishes deep-dives on backend systems design, drawn from real production and portfolio code, on a personal site, dev.to and Substack.

TECHNICAL SKILLS

Languages	Java, Python, JavaScript, TypeScript
Frameworks	Spring Boot, Flask, Fastify
Backend & Architecture	RESTful APIs, microservices, event-driven architecture, pub/sub patterns, synchronous & asynchronous flows, multi-tenancy, stateful vs. stateless systems
Databases & Caching	PostgreSQL, MySQL, MongoDB, DynamoDB, Redis; indexing, query optimisation, keyset pagination, caching strategies
Messaging & Streaming	Familiar with Kafka and RabbitMQ event-driven patterns; publish/subscribe, streaming pipelines, message queuing
Testing	Unit testing, integration testing, TDD, test coverage, regression testing
DevOps & Tooling	Docker, Maven, Git, CI/CD pipelines, build scripting, AWS, Linux
Observability	JVM metrics, health checks, PostHog analytics, log-based debugging, performance monitoring, incident triage
Networking Fundamentals	OSI model, reverse proxies, load balancing, API routing, connection multiplexing, caching layers

EDUCATION

Bachelor of Science, Computer Science — Thika, Kenya

Gretsa University